

Yizheng Xie

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Education

Ph.D. in Computer Science , Distributed Systems, Operating Systems, Cloud Computing Brown University, Providence, RI, USA (Advisor: Nikos Vasilakis)	Sept 2023 – Present
M.S. in Computer Science , Database Systems, Streaming Systems Boston University, Boston, MA, USA (Advisors: Vasiliki Kalavri, John Liagouris)	Sept 2021 – May 2023
B.E. in Electronic Information Engineering , Computer Systems, Machine Learning Chinese University of Hong Kong, Shenzhen, Shenzhen, Guangdong, China	Sept 2017 – May 2021

Working Experience

Meta Platforms , Menlo Park, CA <i>Production Engineer Intern in Kernel Team</i>	May 2022 – Aug 2022
- Improved and integrated Fedora and CentOS Stream forges, issue trackers, and build workflow into Meta internal tools. - Implemented a CLI tool to automate the recursive updating of Rust RPM packages, optimizing parallel chain-build order and ensuring full compatibility (Pagure Repository: pagure.io/fedora-rust/rust-update-set).	

Publications

* Equal contribution.

- [1] **Yizheng Xie***, Evangelos Lamprou*, Jerry Xia*, and Nikos Vasilakis. *Incr: Faster Re-execution via Bolt-on Incrementalization*. In 20th USENIX Symposium on Operating Systems Design and Implementation (**OSDI**), Seattle, WA, July 2026. USENIX Association. ([Paper](#), [Code](#))
- [2] **Yizheng Xie**, Di Jin, Oğuzhan Çölkesen, Vasiliki Kalavri, John Liagouris, and Nikos Vasilakis. *Slowpoke: End-to-end Throughput Optimization Modeling for Microservice Applications*. In 23rd USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Renton, WA, May 2026. USENIX Association. ([Paper](#), [Code](#))
- [3] Zhicheng Huang*, Ramiz Dunder*, **Yizheng Xie**, Konstantinos Kallas, and Nikos Vasilakis. *Fractal: Fault-Tolerant Shell-Script Distribution*. In 23rd USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Renton, WA, May 2026. USENIX Association. ([Paper](#), [Code](#))
- [4] **Yizheng Xie**, Di Jin, and Nikos Vasilakis. *Towards Hybrid Cooperative-Preemptive Scheduling*. In Proceedings of the 13th Workshop on Programming Languages and Operating Systems (**PLOS**), Seoul, Republic of Korea, October 2025. ACM. ([Paper](#))

Research

Slowpoke: End-to-end Throughput Optimization Modeling for Microservices <i>To appear at NSDI'26</i>	Sept 2023 – Apr 2025
- Proposed a novel linear programming-based performance model to quantify end-to-end throughput improvements of hypothetical microservice optimizations without requiring white-box knowledge. - Developed the first system to accurately predict throughput in complex microservices, achieving RMSE of 2.1% across several real-world benchmarks and 1.9% across 108 synthetic benchmarks with diverse characteristics.	
Fractal: Fault-Tolerant Shell-Script Distribution <i>To appear at NSDI'26</i>	Sept 2023 – Dec 2024
- Developed a system for fault-tolerant distribution of unmodified shell scripts, delivering average speedups of $9.6\times$ over Bash, $5.5\times$ over Hadoop Streaming (AHS), and $1.2\times$ over DiSH, the state-of-the-art shell scaleout system. - Designed a lightweight fault-recovery mechanism—including remote-pipe instrumentation and dynamic result persistence—that achieves recovery within $1.26\times$ of fault-free execution and delivers a $9.3\times$ speedup over AHS.	
Incr: Faster Incremental Execution, Opaquely <i>To appear at OSDI'26</i>	Aug 2025 – Jan 2026
Developing a runtime system that automatically memoizes transient results and traces system calls to capture fine-grained and side-effectful dependencies, enabling orders-of-magnitude faster incremental development while providing correctness guarantees.	
leash: Scaling Out Shell Scripts with Resumable Cloud Computing <i>Under submission (SOSP'26), with Nikos Pagonas, Haoran Zhang and Konstantinos Kallas</i>	Sept 2023 – Present

- Developed a system to scale out unmodified shell scripts using cloud (serverful and serverless) computing, achieving $12.7\times$ average and $154.2\times$ peak speedups over Bash; in some cases both faster and cheaper, e.g., achieving a $6.5\times$ speedup while reducing cost by 16.4% compared to PASH—the state-of-the-art shell parallelization system.
- Designed a *suspend-and-resume* protocol that extends AWS Lambda beyond the 15-minute execution limit while preserving correctness and Unix-style streaming efficiency, with negligible slowdown (avg: 3%).

MeshBench: Benchmarking and Analyzing Service Meshes for Microservices

Aug 2025 – Present

In preparation (ATC'26), with Alice Song, Deepti Raghavan and Akshay Narayan

- Developing a comprehensive benchmarking suite to evaluate the performance overhead and trade-offs of leading service meshes—e.g., Istio, Cilium, Ambient, Canal—in microservice environments, and analyzing how service mesh architectures and functionalities, e.g., security and observability, affect performance and resource utilization.

Invited Talk

Towards Automated Performance Optimizations for Service-Oriented Architectures

Oct 2025

Boston University, Boston, MA, USA

Teaching

CSCI 1380 Distributed Systems, Brown University, *Teaching Assistant (Spring 2023, Spring 2024)*

- Co-designed and implemented hands-on assignments for building a large-scale search engine that emphasized core distributed systems concepts, including a distributed key-value store for scalable data storage, data sharding for load balancing, consensus protocols for data consistency, and a MapReduce system for efficient data processing.

CSCI 2952R Systems Transforming Systems Seminar, Brown University, *Teaching Assistant (Fall 2024)*

- Mentored students in conducting state-of-the-art systems research across operating systems, distributed systems, and cloud computing, guiding them through system implementation, experimental evaluation, and academic paper writing.

Skills

- **Programming Languages:** Go, Python, C/C++, Rust, Java, JavaScript, Shell, SQL
- **Systems:** Kubernetes, Docker, Istio, AWS (e.g., Lambda), Linux/Unix, Apache Flink, Spark, V8/libUV, MongoDB, Redis
- **Tools:** Git, CI/CD (GitHub Actions, GitLab CI), Helm, Make, CMake, LaTeX, Vim, perf

Languages and Interests

- **Languages:** Mandarin (native), English (fluent)
- **Interests:** Photography, Video Editing, Badminton, Basketball, Traveling, East Asian History